

Exam Date & Time: 17-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

DISTRIBUTED CLOUD SECURITY [IT-4002]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A fintech company has implemented Multi-Factor Authentication (MFA) to secure access to its financial application, which is used by employees to manage sensitive client data. (5)
 - A) Apply your understanding of Multi-Factor Authentication (MFA) to explain how each of the factors can be implemented in this scenario to strengthen login security.
 - B) A government defense agency uses a secure communication application for agents to access classified data across a distributed environment. To protect against unauthorized access and replay attacks, the system implements the Denning Protocol, which relies on timestamps and symmetric encryption. Apply the Denning Protocol to this distributed environment to demonstrate how secure communication is established between agents and servers. Illustrate how the use of timestamps and symmetric encryption in the protocol helps mitigate the risk of replay attacks in this setting. (5)
- 2) A large healthcare organization stores sensitive patient records across its digital systems. Due to the absence of a structured access control policy, employees from various departments can access more information than necessary, which recently resulted in an accidental data leak. (5)
 - A) How would you apply Role-Based Access Control (RBAC) to prevent such incidents in this organization? Discuss the key principles, benefits, and implementation challenges of RBAC system for a healthcare environment.
 - B) Illustrate how IAM operates with a neat sketch and provide a real-world scenario demonstrating its role in enhancing security and compliance in an organization. (5)
- 3) How would you apply Zero Trust Network Access (ZTNA) in place of traditional VPNs to improve secure remote access in your organization? Explain how this approach addresses insider threats, integrates with existing cybersecurity frameworks, and overcomes potential implementation challenges. (5)
 - A)
 - B) A financial institution experiences frequent cyberattacks targeting both its internal servers and public-facing applications. The security team is tasked with enhancing intrusion detection to minimize threats across the network and endpoints. Apply your knowledge of NIDS and HIDS to recommend an effective intrusion detection strategy for the institution. Justify your choice by comparing the strengths and limitations of each system. (5)

- 4) Apply security best practices to address the major challenges in cloud computing and explain how organizations can implement effective strategies to ensure data protection, regulatory compliance, and system integrity. (5)
- A)
- B) Apply the CIA Risk Assessment Framework to evaluate and address organizational risks related to Confidentiality, Integrity, and Availability, and justify how this approach supports overall information security. (5)
- 5) An organization is planning to virtualize its data centre to optimize resource usage and improve scalability. As part of the IT team, you are responsible for evaluating different server virtualization approaches to determine the best fit. Analyse the main types of server virtualization by comparing their architectures, functionalities, and security implications. How do these differences influence the selection of an appropriate virtualization method based on the organization's needs? (5)
- A)
- B) Your organization is transitioning from a traditional physical server environment to a virtualized infrastructure using Microsoft Hyper-V. As a system administrator, you are required to evaluate its suitability for deployment. Analyse the architecture of Microsoft Hyper-V by identifying its key components, core functionalities, and built-in security features, and explain how they support efficient and secure virtualization in an enterprise setting. (5)

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